

FreshFlow Beverages Pvt. Ltd.

FreshFlow Bottling Plant - Pune Production Facility
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STANDARD OPERATING PROCEDURE

Sanitization Procedure for Utilities Line 68

Department: Utilities

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Prepared by: Engineering / QA Department
Approved by: Plant Manager

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Prepared By	Employee 14
Reviewed By	Anita Desai
Approved By	Rajesh Kulkarni - Plant Manager
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1. Purpose

Safe operation of plant utilities (compressed air, steam, chilled water, CO2, electrical) to support continuous production.

2. Scope

Utilities department operators, Maintenance engineers.

3. Prerequisites and Requirements

3.1 Training Requirements

- Boiler operator certification (IBR)
- Compressor operation training
- Electrical safety training (LV)

3.2 Tools and Materials Required

- Calibrated instruments as listed in the plant calibration schedule.
- Appropriate PPE as specified in Section 4.
- Completed permit-to-work (if required for maintenance activities).
- Access to relevant equipment manual and P&ID drawings.
- CMMS access to create and close work orders.

4. Health, Safety and Environmental Requirements

! WARNING

This SOP must not be carried out without appropriate PPE and, where required, a valid Permit to Work (SOP-SAF-006). If in doubt about safety, STOP and consult the Safety Officer (EMP009 / EMP017).

4.1 Identified Hazards

- High-pressure steam (7 kg/cm²) - scald and explosion risk
- High-voltage electrical (415V / 11kV) - electrocution risk
- CO₂ stored at high pressure (-28°C liquid) - cryogenic and pressure hazard
- Diesel generator - hot exhaust, rotating parts, high voltage output

4.2 Emergency Response

In case of any emergency during this procedure: Press the nearest Emergency Stop, alert personnel in the vicinity, call the Safety Officer and Shift Supervisor. Refer to the Emergency Response Plan (SOP-SAF-004) and site evacuation procedure.

5. Detailed Procedure

5.1 Pre-Task Verification

1. Confirm the relevant equipment is available and no conflicting activities are planned.
2. Check shift handover log for any outstanding issues with this equipment.
3. Verify all required materials, tools, and chemicals are available and correctly labelled.
4. Obtain all required permits (PTW, hot work, confined space as applicable).
5. Brief all involved personnel on this SOP before starting.

5.2 Main Procedure

1. Confirm production has stopped and product has been drained from the system.
2. Ensure CIP supply pump P301 is ready and CIP tanks have correct chemical charge.
3. Connect CIP return line and verify all CIP circuit valves are in correct position.
4. Start pre-rinse with RO water for 5 minutes - verify runoff is clear.
5. Switch to caustic recirculation: 2% NaOH at 75°C for 20 minutes minimum.
6. Monitor caustic concentration with inline conductivity meter - maintain 15-20 mS/cm.
7. Flush with RO water until conductivity <100 uS/cm and pH 6.5-7.5.
8. Perform acid circulation: 1% HNO₃ at 65°C for 15 minutes (passivation step).
9. Final rinse with RO water until conductivity <20 uS/cm and pH 6.5-7.5.
10. Take microbiological swab from defined sample points. Record in QC log.
11. Equipment is released for production only after QC verification (Form FFB-QA-CIP-VER).

6. Quality and Verification Checks

Verification Check	When	Responsible	Record
All product contact surfaces visually clean	Before production	Operator	Shift Log
CIP conductivity <=20 uS/cm	After each CIP	Operator	CIP Record
CIP pH 6.5-7.5	After each CIP	Operator	CIP Record
Microbiological swab <10 CFU/cm2	After each CIP	QC Lab	Micro Report
Key process parameters in normal range	At startup	Operator	SCADA Historian
First-off product sample - Brix, pH, CO2	After startup	QC Analyst	QC Lab Report

7. Non-Conformance and Escalation

If any check fails or if the procedure cannot be completed as written: STOP the task. Do not force equipment to operate outside specification. Escalate to the Shift Supervisor (Suresh Patil / Ravi Deshpande). Raise a Non-Conformance Report (NCR) in the CMMS system. QC Manager (Anita Desai) must be notified for any product safety impact.